BCS-054	SOFT COMPUTING	L T P 3 0 0	3 Credits

Course Objectives:

CO1	Introduce students to fuzzy systems, fuzzy logic and its applications.
CO2	Explain the students about Artificial Neural Networks and various categories of ANN.
CO3	To learn the concept of Neuro Fuzzy Modelling
CO4	To understand Genetic algorithm
CO5	Introduction to MATLAB and various commands

Detailed Syllabus

MODULE-I

Artificial Neural Networks: Basic concepts - Single layer perception - Multilayer Perception - Supervised and Unsupervised learning – Back propagation networks - Kohnen's self organizing networks - Hopfield network.

Fuzzy Systems: Fuzzy sets, Fuzzy Relations and Fuzzy reasoning, Fuzzy functions - Decomposition – Fuzzy automata and languages - Fuzzy control methods - Fuzzy decision making.

MODULE-II

Neuro-Fuzzy Modeling: Adaptive networks based Fuzzy interface systems - Classification and Regression Trees –Data clustering algorithms - Rule based structure identification - Neuro-Fuzzy controls –Simulated annealing – Evolutionary computation.

Genetic Algorithms: Survival of the Fittest - Fitness Computations - Cross over - Mutation - Reproduction –Rankmethod - Rank space method.

MODULE-III

Application of Soft Computing: Optimiation of traveling salesman problem using Genetic Algorithm, Genetic algorithm basedInternet Search Techniques, Soft computing based hybrid fuzzy controller, IntoductiontoMATLAB Environment for Soft computing Techniques.

Text Books:

1. Sivanandam, Deepa, " Principles of Soft Computing", Wiley
2. Jang J.S.R, Sun C.T. and Mizutani E, "Neuro-Fuzzy and Soft computing", PrenticeHall
3. Timothy J. Ross, "Fuzzy Logic with Engineering Applications", McGraw Hill

Reference Books:

1. LaureneFausett, "Fundamentals of Neural Networks", Prentice Hall
2. D.E. Goldberg, "Genetic Algorithms: Search, Optimization and Machine Learning",Addison Wesley
3. Wang, “Fuzzy Logic”, Springer

Course Outcomes: After the completion of the course the student will be able to:

CO1	Learn about soft computing techniques and their applications.
CO2	Analyze various neural network architectures.
CO3	Define the fuzzy systems.
CO4	Understand the genetic algorithm concepts and their applications.
CO5	Identify and select a suitable Soft Computing technology to solve the problem; construct a solution and implement a Soft Computing solution. .